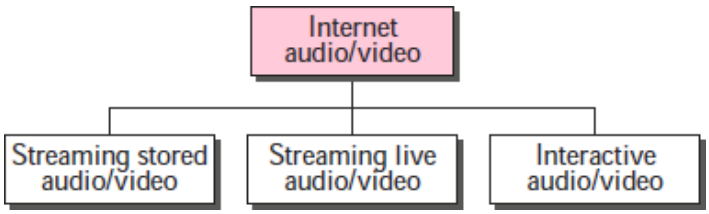
4.1. Multimedia in the Internet: Streaming stored audio/video, streaming live audio/video

INTRODUCTION

We can divide audio and video services into three broad categories: **streaming stored audio/video, streaming live audio/video,** and **interactive audio/video**. Streaming means a user can listen (or watch)



the file after the downloading has started.

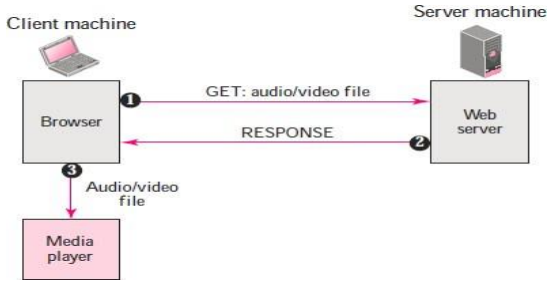
In the **first category**, streaming stored audio/video, **the files are compressed and stored on a server**. A client downloads the files through the Internet. This is sometimes referred to as **on-demand audio/video.** In the **second category**, streaming live audio/video **refers to the broadcasting of radio and TV programs through the Internet.** In the **third category**, interactive audio/video **refers to the use of the Internet for interactive audio/video applications.** A good example of this application is Internet telephony and Internet teleconferencing.

1. STREAMING STORED AUDIO/VIDEO

Downloading these types of files from a server can be different from downloading other types of files.

First Approach: Using a Web Server

A compressed audio/video file can be downloaded as a text file. The **client (browser)** can use the services of **HTTP** and send a GET message to download the file. The **Web server** can send the compressed file to the browser. The browser can then use a help application, normally called a **media**



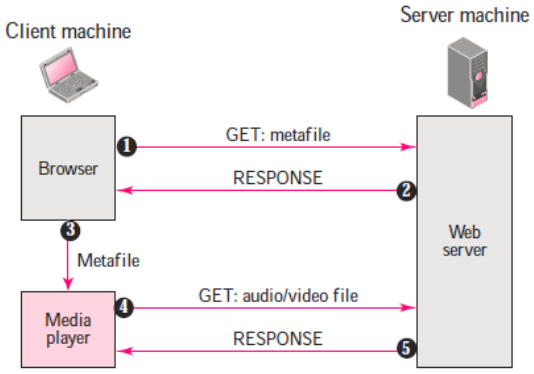
player, to play the file. The file needs to download completely before it can be played.

Second Approach: Using a Web Server with Metafile

In another approach, the media player is directly connected to the Web server for downloading the audio/video file. The Web server stores two files: **the actual audio/video file** and a **metafile** that holds

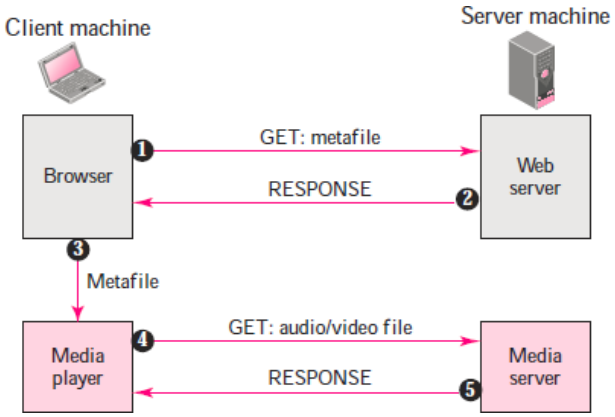
information about the audio/video file.

1. The HTTP client accesses the Web server using the GET message.
2. The information about the metafile comes in the response.
3. The metafile is passed to the media player.
4. The media player uses the URL in the metafile to access the audio/video file.
5. The Web server responds.



Third Approach: Using a Media Server

The **problem** with the **second approach** is that **the browser and the media player both use the services of HTTP**. HTTP is designed to run over TCP. This is appropriate for retrieving the metafile, but not for retrieving the audio/video file**. The reason is that TCP retransmits a lost or damaged segment, which is counter to the philosophy of streaming.** We need to dismiss TCP and its error control; we need to use UDP. However, HTTP, which accesses the Web server, and the Web server itself are designed for TCP; we need another server, a **media server.**

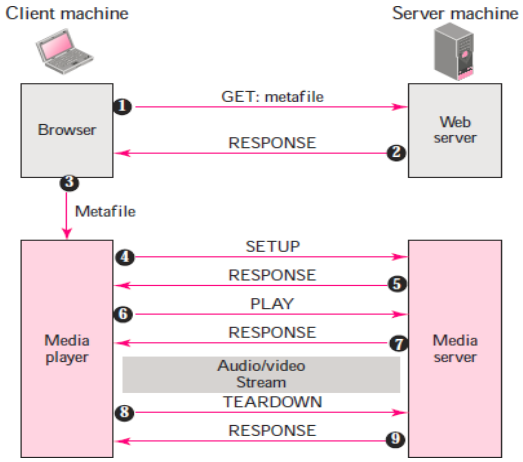


1. The HTTP client accesses the Web server using a GET message.
2. The information about the metafile comes in the response.
3. The metafile is passed to the media player.

4. The media player uses the URL in the metafile to access the media server to download the file. Downloading can take place by any protocol that uses UDP.
5. The media server responds.

Fourth Approach: Using a Media Server and RTSP

The **Real-Time Streaming Protocol (RTSP)** is a control protocol designed to add more functionalities to the streaming process. Using RTSP, we can control the playing of audio/video. Figure 5 shows a media server and RTSP.



1. The HTTP client accesses the Web server using a GET message.
2. The information about the metafile comes in the response.
3. The metafile is passed to the media player.
4. The media player sends a SETUP message to create a connection with the media server.
5. The media server responds.
6. The media player sends a PLAY message to start playing (downloading).
7. The audio/video file is downloaded using another protocol that runs over UDP.
8. The connection is broken using the TEARDOWN message.
9. The media server responds.

2. STREAMING LIVE AUDIO/VIDEO

Streaming live audio/video is similar to the broadcasting of audio and video by radio and TV stations. Instead of broadcasting to the air, the stations broadcast through the Internet. There are several similarities between streaming stored audio/video and streaming live audio/video. They are both

sensitive to delay; neither can accept retransmission. However, there is a difference. In the first application, the communication is unicast and on-demand. In the second, the communication is multicast and live. Live streaming is better suited to the multicast services of IP and the use ofprotocols such as UDP and RTP.

Examples: Internet Radio, Internet Television (ITV), Internet protocol television (IPTV)

29-7 RTCP

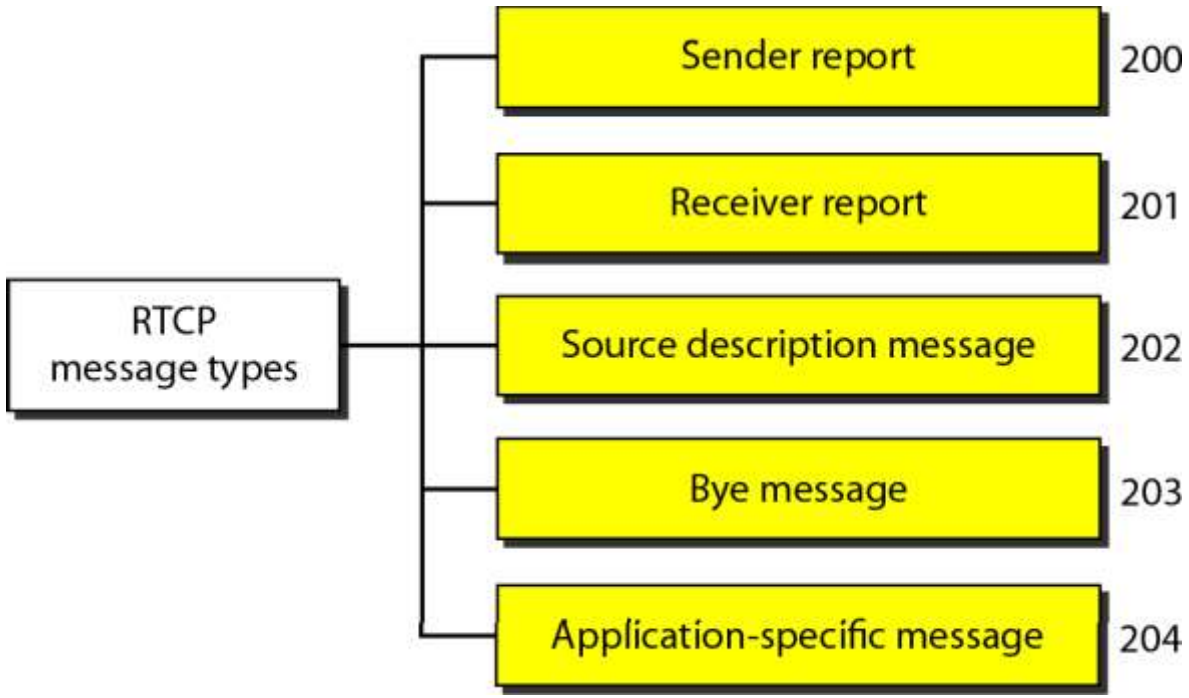
RTP allows only one type of message, one that carries data from the source to the destination. In many cases, there is a need for other messages in a session.... These messages control the flow and quality of data and allow the recipient to send feedback to the source

Or sources.Real-time Transport Control Protocol (RTCP)

RTCP has five types of messages, as shown in Figure.

The number next to each box defines the type of the message.

Figure ***RTCP message types***



1.Sender Report

The sender report is sent periodically by the **active senders** in a conference to **report transmission and reception** statistics for all RTP packets sent during the interval.

The sender report includes an **absolute timestamp**, which is the number of seconds elapsed

The absolute timestamp **allows the receiver to synchronize different RTP messages.**

It is particularly **important when both audio and video are transmitted** (audio and video transmissions use separate relative timestamps).

Receiver Report

The receiver report is for **passive participants**, those that do not send RTP packets.
The report informs the sender and other receivers **about the quality of service.**

Source Description Message

The source **periodically** sends a source description message to give **additional information about itself**.
This information can be the name, e-mail address, telephone number,and address of the owner or controller of the source.

Bye Message

A **source** sends a bye message to **shut down a stream**.

It allows the source to **announce that it is leaving the conference.**

Although other sources can detect the absence of a source, **this message is a direct announcement.**

Application-Specific Message

- The application-specific message is a **packet for an application that wants to use new applications** (not defined in the standard).
  It allows the **definition of a new message Type.**

UDPPort

RTCP, like RTP, **does not use a well-known UDP port**.
It **uses a temporary port**.
The UDP port chosen **must be the number immediately following the UDP port selected for RTP.**
It must be an **odd-numbered port**.

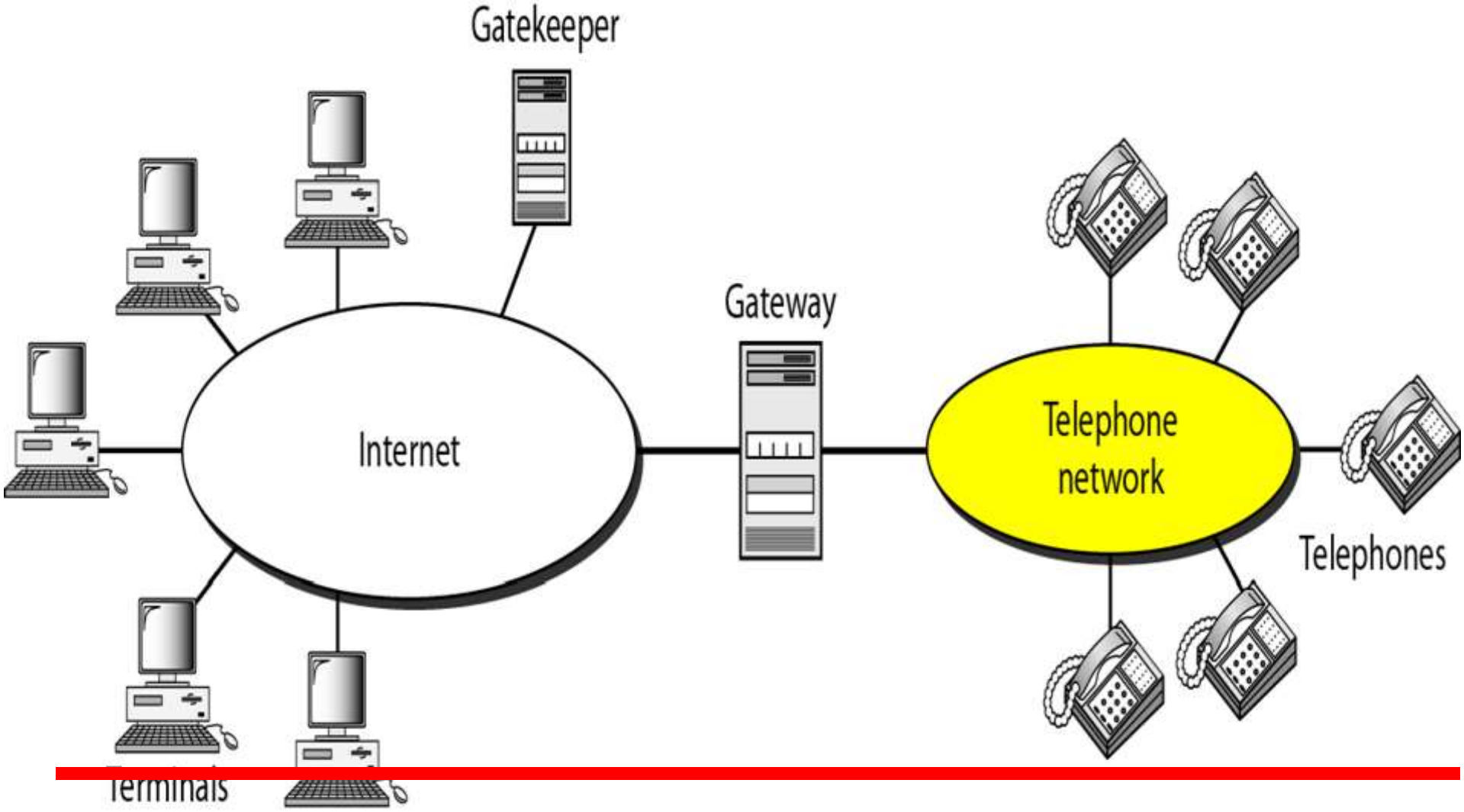
Note

RTCP uses an odd-numbered UDP port number that follows the port number selected for RTP.

H.323

H.323 is a standard designed by ITV to **allow telephones on the public telephone network to talk to computers** (called **terminals** in H.323) connected to the Internet.

H.323 architecture



- **A gateway** connects the Internet to the telephone network.
- In general, a gateway is a five-layer device that can **translate a message from one protocol stack to another**.

-The gateway here does exactly the same thing. **It transforms a telephone network message to an Internet message.**

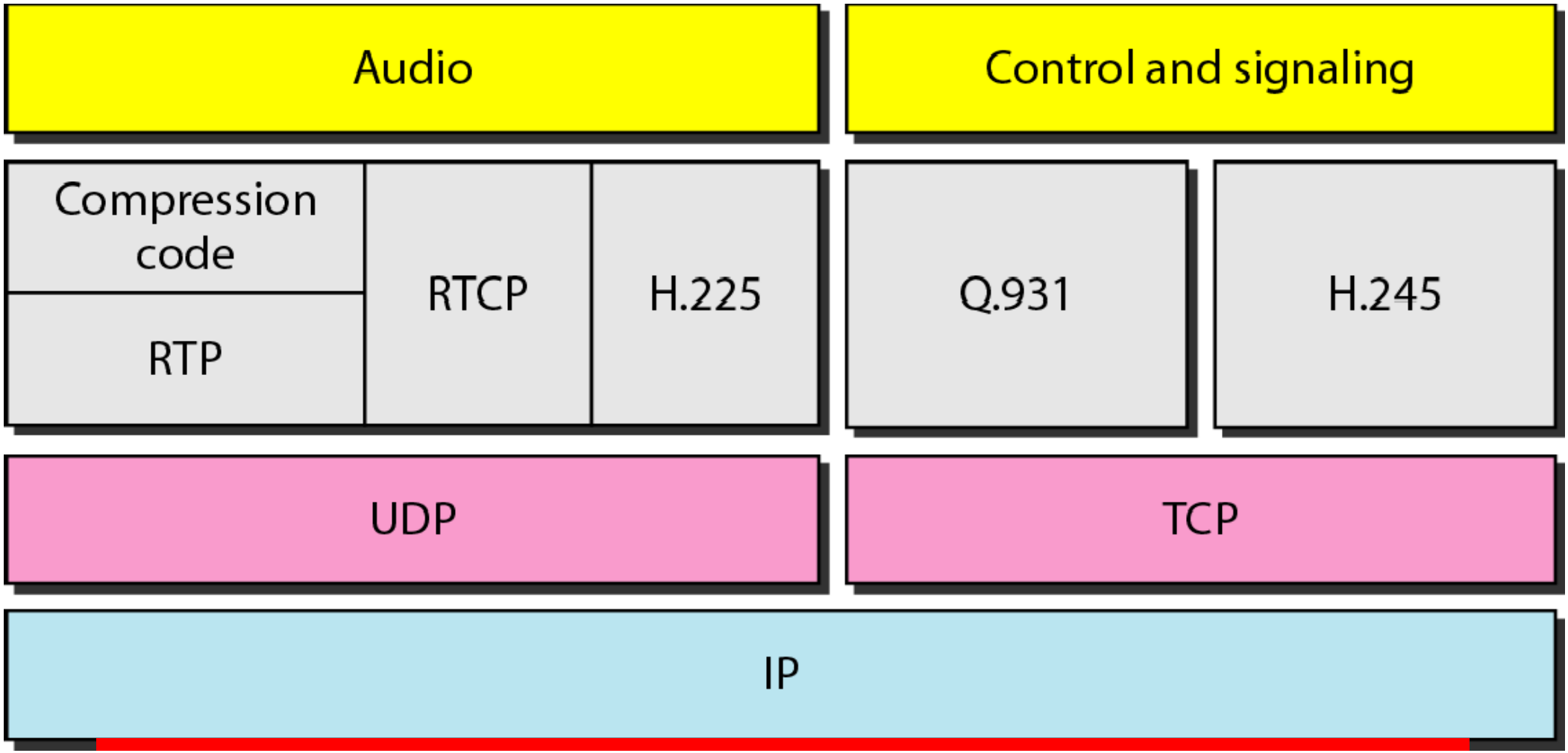
- The **gatekeeper** server on the local area network plays the role of the **registrar server**

Protocols

- H.323 uses a number of protocols to **establish and maintain voice (or video) communication**.
- H.323 uses **G.71 or G.723.1 for compression**.
- It uses a protocol named **H.245 which allows the parties to negotiate the compression method**.
- Protocol **Q.931 is used for establishing and terminating connections.**
- Another protocol called **H.225, or RAS (Registration!Administration!Status), is used for registration with the gatekeeper.**

  -to negotiate bandwidth

H.323 protocols

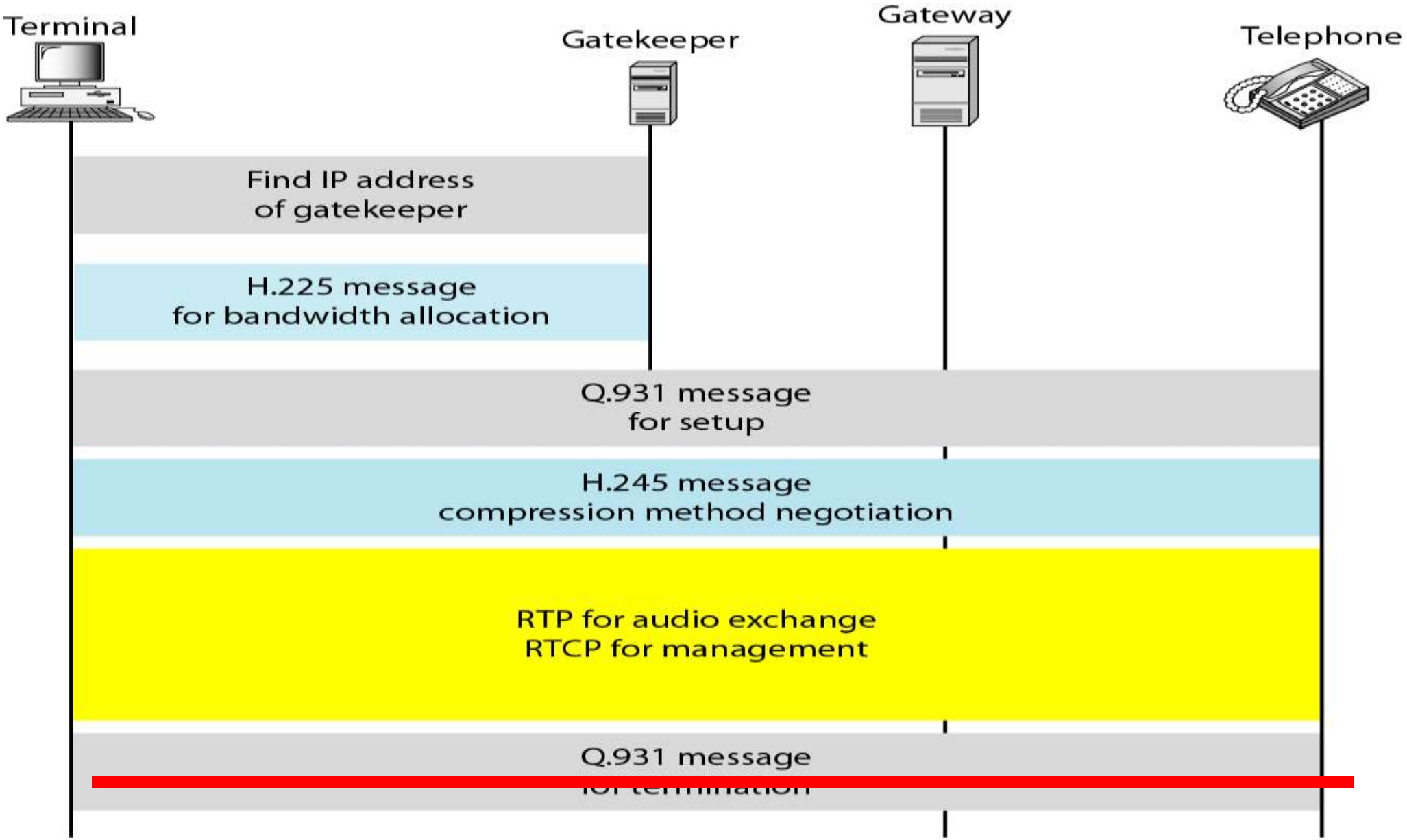


Operation

- Let us show the operation of a telephone communication using H.323 with a simple example.
- Figure shows the **steps used by a terminal to communicate with aTelephone.**

  1. The **terminal sends a broadcast message to the gatekeeper**. The gatekeeper responds with its IP address.
  2. The terminal and gatekeeper communicate, using H.225 to **negotiate bandwidth**.
  3. The terminal, gatekeeper, gateway, and telephone communicate by **using Q.931 to set up a connection**.

Figure 29.27 *H.323 example*



4. The terminal, gatekeeper, gateway, and telephone communicate by using **H.245 to negotiate the compression method.**

5. The terminal, gateway, and telephone **exchange audio by using RTP under the management of RTCP.**

6. The terminal, gatekeeper, gateway, and telephone communicate by using **Q.931 to terminate the communication.**

REAL TIME INTERACTIVE PROTOCOLS: SIP, SCTP

SESSION INITIATION PROTOCOL (SIP)

Voice over IP

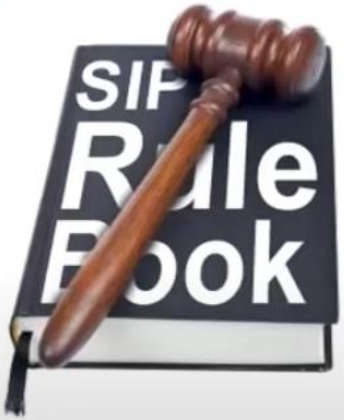




IP Network







The Session Initiation Protocol

- How to locate the other party
- Which codec to use
- How to construct & send IP signalling packets
- How to act. What to do to manage the communication

SESSION INITIATION PROTOCOL

- SIP is a **signaling protocol** used to create, modify, and terminate a multimedia session over the Internet Protocol. A **session** is nothing but a simple call between two endpoints.
- An **endpoint** can be a smartphone, a laptop, or any device that can receive and send multimedia content over the Internet.
- SIP is an **application layer protocol** defined by IETF (Internet Engineering Task Force) standard. It is defined in **RFC 3261**.

- It provides mechanisms for **establishing calls between a caller and a callee over an IP network**.

  -It allows the caller to **notify** the callee that it wants to start a call.

  -It allows the participants to agree on **encodings**

. -It also allows participants to **end calls**.

- Other **SIP applications** include file transfer, instant messaging, video conferencing, online games, and steaming multimedia distribution.

SIP ADDRESS

In a **regular telephone communication**, a phone number identifies the sender and another phone number identifies the receiver.

In **SIP**, the sender and receiver are often identified by any of these.

1.An Email address.

2.An IP address.

3.A Phone number



SIP MESSAGES

- SIP may be a **text based protocol modeled on HTTP**.
- It **uses messages in ASCII text**. Each message features **a header and a body**. the varied SIP messages are as follows.

Sr. No.	Message	Description
1.	INVITE	It requests for initiation of a session.
2.	ACK	It confirms that session has initiated.
3.	BYE	It request for the termination of the session.
4.	OPTIONS	It query a host about its capabilities.
5.	CANCEL	It will cancel the pending request.
6.	REGISTER	It informs a redirection server about the user's current location.

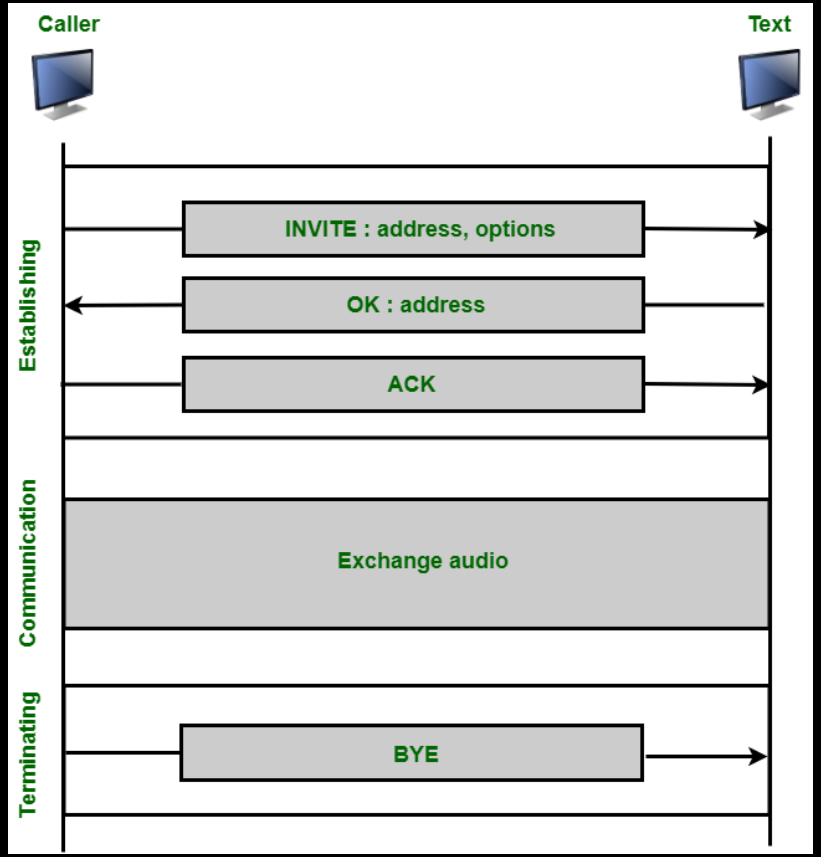
SIP SESSION

- A simple session using SIP consists of the following.

1.Establishing a session.

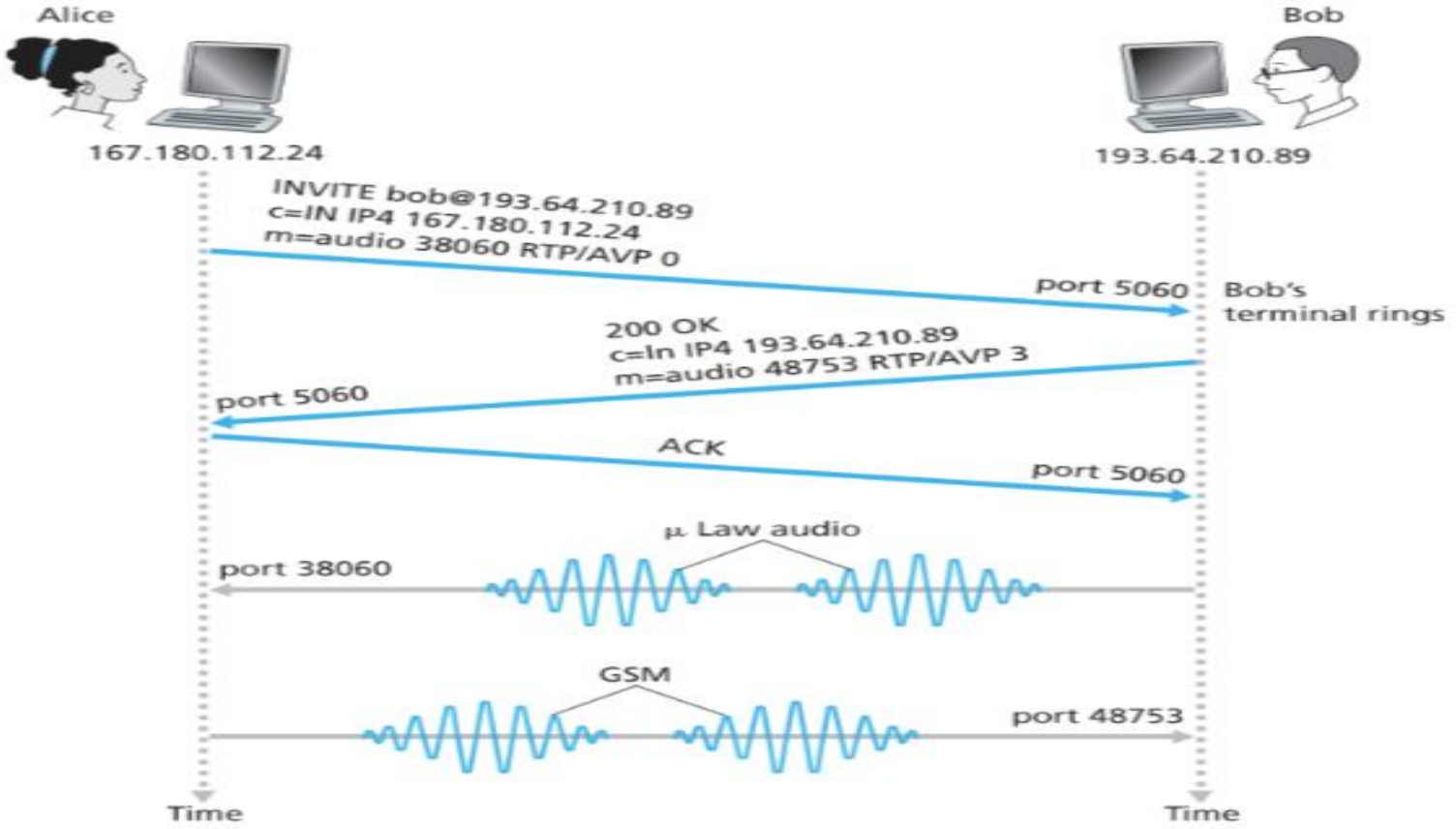
2.Communication.

3.Terminating the session



SIP SESSION

- **Establishing a session :**It requires a three-way handshake. The caller will send INVITE message. If the callee is willing to start out, he/she sends a reply message. to verify that a reply code is received, the caller send an ACK message.
- **Communication :**After establishment of session, the caller and callee communicate using two temporary ports.
- **Terminating the session :**The session can often terminated by using BYE message send by either caller or callee.


Alice
Bob
167.180.112.24
193.64.210.89
INVITE bob@193.64.210.89
c=IN IP4 167.180.112.24
m=audio 38060 RTP/AVP 0
port 5060
Bob's terminal rings
200 OK
c=In IP4 193.64.210.89
m=audio 48753 RTP/AVP 3
port 5060
ACK
port 5060
μ Law audio
port 38060
GSM
port 48753
Time
Time

SIP (EXAMPLE) - **SIP CALL-ESTABLISHMENT PROCESS:**

Initiation:

•Alice wants to call Bob using SIP.

•Both Alice and Bob have SIP-based software on their PCs.

INVITE Message:

•Alice sends an INVITE message to Bob's IP address (193.64.210.89) on SIP port 5060.

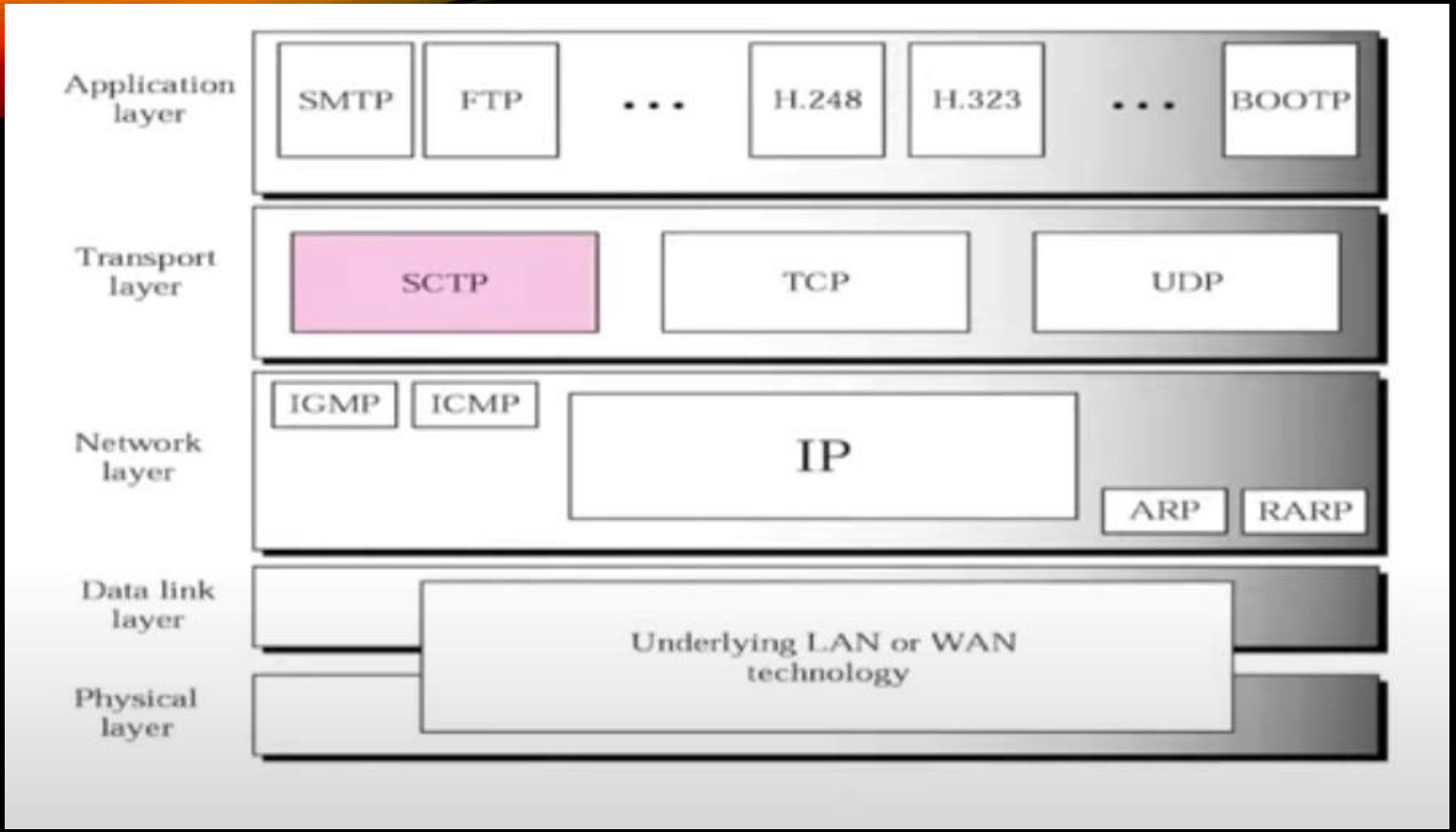
•INVITE includes identifiers, IP addresses, audio format preferences (AVP 0 for Alice, PCM μ-law for Bob), and desired RTP ports.

- **Response from Bob:**
- Bob responds with a OK message, indicating acceptance and providing his IP address, audio preferences (PCM μ-law for Bob), and RTP port.
- **Acknowledgment:** Alice sends a SIP acknowledgment message to confirm the acceptance.
- **Audio Transmission:** Bob and Alice can now talk. Bob encodes and packetizes audio as requested, sending to Alice's IP address and port. Alice does the same, sending audio to Bob's IP address and port.

SIP (EXAMPLE) - **SIP CALL-ESTABLISHMENT PROCESS:**

- **HANDLING CODEC MISMATCH:**
- If Bob lacks the required audio codec (PCM μ-law), he responds with a 606 Not Acceptable, listing available codecs.
- Alice can choose an alternative codec and send another INVITE message.
- Bob may reject the call with various rejection reply codes ("busy," "gone," "payment required," "forbidden").

STREAM CONTROL TRANSMISSION PROTOCOL. (SCTP)


Application layer
SMTP
FTP
. . .
H.248
H.323
. . .
BOOTP
Transport layer
SCTP
TCP
UDP
Network layer
IGMP
ICMP
IP
ARP
RARP
Data link layer
Underlying LAN or WAN technology
Physical layer

TCP/IP

UDP, TCP, SCTP

TCP

- Byte-oriented protocol
- Receives a message, store them as a stream of bytes and sends them in segments
- Reliable
- Has congestion control and flow control

- UDP: Message-oriented
- A process delivers a message to UDP
- UDP conserve the message boundaries
- (Each message is independent from each other)
- Good for real-time or IP telephony
- Unreliable
- (Lack congestion control and flow control)
- Bad for real-time or IP telephony

SCTP

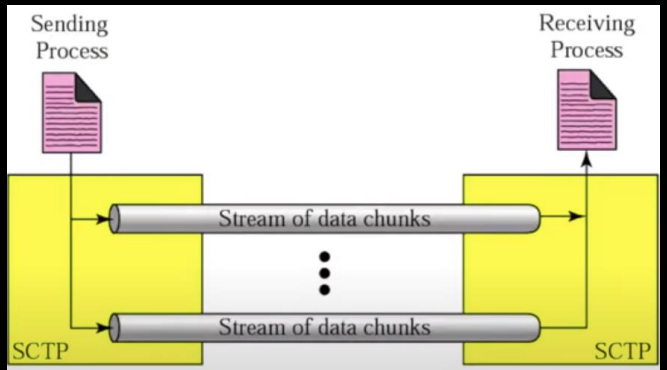
- Combine the features of UDP and TCP
- Reliable message-oriented protocol
- Preserve the message boundaries
- Other innovative features
  - Association, Data transfer/Delivery
  - Fragmentation, Error/Congestion Control

SCTP

- SCTP stands for **Stream Control Transmission Protocol**.
- It is a **connection- oriented protocol** in computer networks which provides a **full-duplex** association i.e., **transmitting multiple streams of data between two end points** at the same time that have established a connection in network.
- It is sometimes referred to as **next generation TCP** , SCTP makes it easier to support **telephonic conversation on Internet**.
- A telephonic conversation requires transmitting of **voice along with other data at the same time on both ends**, SCTP protocol makes it easier to establish reliable connection.

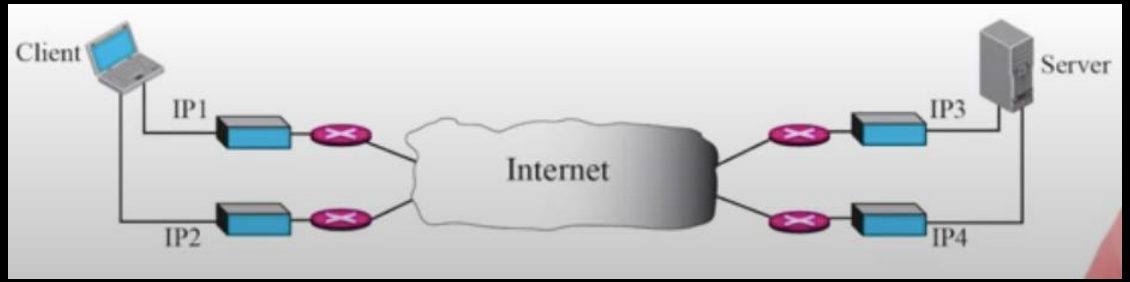
KEY FEATURES OF SCTP:

- **Connection-Oriented:** SCTP is connection-oriented, like TCP, ensuring reliable and ordered delivery of messages between endpoints. However, unlike TCP, it supports **multiple streams** within a single connection.
- **Multi-Stream Support:** One of the distinguishing features of SCTP is its ability to transmit multiple streams of data in parallel within a single association (connection). Each stream operates independently, allowing for **more efficient and flexible** data transfer.



KEY FEATURES OF SCTP:

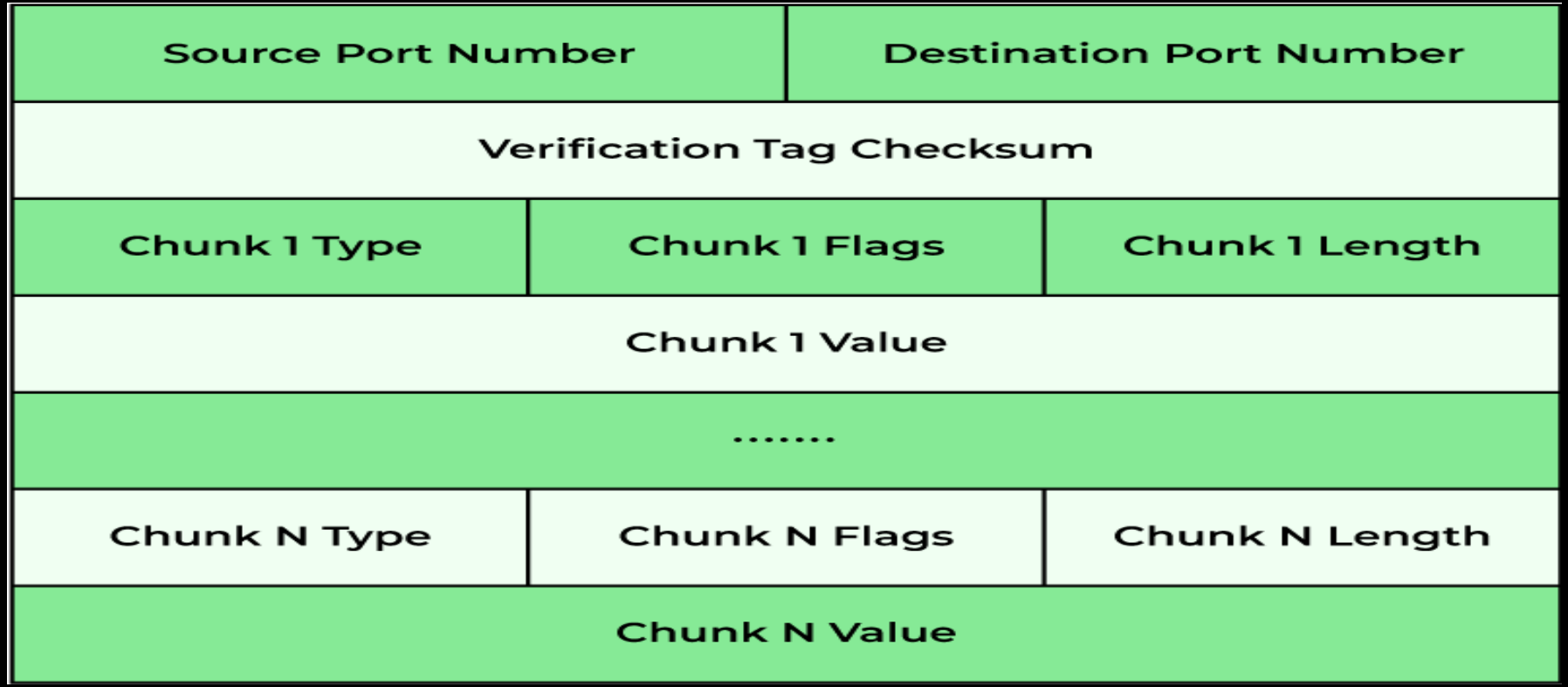
- **Message-Oriented:** SCTP is message-oriented, preserving **message boundaries**. This means that the boundaries of application messages are maintained during transmission, which is particularly useful for applications that send **discrete messages**.
- **Multi-Homing:** SCTP supports multi-homing**, allowing endpoints to have multiple IP addresses**. This feature **enhances fault tolerance and resilienc**e by providing alternative paths for data transmission if one network path fails.



KEY FEATURES OF SCTP:

- **Reliability:** SCTP ensures reliable data delivery by using **acknowledgments and retransmissions**. It includes mechanisms to **detect and handle network failures**.
- **Ordered and Unordered Delivery:** SCTP allows for both ordered and unordered delivery of data. Ordered delivery ensures that **messages are received in the same order** they were sent, while unordered delivery allows for more flexibility in receiving **messages out of order.**
- **Flow Control:** SCTP includes built-in flow control mechanisms to prevent overwhelming a receiver with data. It adapts to the receiver's processing capabilities.

SCTP PACKET STRUCTURE OVERVIEW



SCTP PACKET - COMMON HEADER SECTION

Field	Description
Source port number	Identifies the sending port.
Destination port number	Identifies the receiving port. The hosts use the destination port number to route the packet to the appropriate destination or an application.
Verification tag	Distinguishes stale packets from a previous connection. This is a 32-bit random value created during initialization.
Checksum	Uses the cyclic redundancy check (CRC32) algorithm to detect errors that might have been introduced during data transmission.

SCTP PACKET - DATA CHUNK SECTION

Field	Description
Chunk Type	Identifies the contents of the chunk value field. This is 1- byte long.
Chunk Flags	Consists of 8 flag-bits whose definition varies with the chunk type. The default value is zero. This indicates that no application identifier is specified by the upper layer for the data.
Chunk Length	Specifies the total length of the chunk in bytes. This field is 2 - bytes long. If the chunk does not form a multiple of 4 bytes (that is, the length is not a multiple of 4) it is implicitly padded with zeros which are not included in the chunk length.
Chunk Value	A general purpose data field.

APPLICATIONS

- **Telecommunication Networks:** SCTP is used in various telecommunication applications, such as **signaling and call setup in the SS7 network.**
- **Voice over IP (VoIP):** SCTP can be employed in VoIP applications, providing a reliable and efficient transport layer for voice communication.
- **WebRTC:** SCTP is used in conjunction with Web Real-Time Communication (WebRTC) to facilitate **real-time communication in web browsers.**
- **File Transfer:** SCTP can be utilized for reliable file transfer applications, particularly when **multi-streaming and ordered delivery are important**.

AUDIO & IMAGE COMPRESSION

COMPRESSION

- Compression is a process by which we reduce data size by ***reducting the number of bits*** needed to represent data.
- Compressing data can ***save storage capacity, speed file transfer, and decrease costs for storage hardware and network bandwidth.***
- Compression is performed by ***program that uses a formula or algorithm*** to determine how to shrink the size of the data.

AUDIO COMPRESSION

- Audio signals are ***digitized using 8/16 bits per sample and compressed using either PCM or Adaptive Differential Pulse Code Modulation(ADPCM) technique.***
- In this a ***reduction to around 4 bits per sample*** is achieved at quality
- Digital audio compression ***allows the efficient storage and transmission of audio data***.
- The ***various audio compression techniques*** offer different levels of complexity, compressed audio quality and amount of data compression.

AUDIO COMPRESSION

- A ***sample of sound is taken and stored as digital information in bits and bytes***.
- The ***sample size are either 8 bits or 16 bits***.
- The value of each sample is rounded off to the nearest integer known as ***quantization***.

AUDIO COMPRESSION

- Audio compression is a form of data compression designed to ***reduce the size of audio data files***.
- Audio compression can mean two things:
  - Audio data compression.
  - Audio level compression.

AUDIO DATA COMPRESSION

- Audio data compression in which ***the amount of data in a recorded waveform is reduced*** for transmission.
- This is ***used in MP3 encoding, internet radio*** etc.
- It can be either:

  - Lossless Audio compression.
  - Lossy Audio compression.

AUDIO DATA COMPRESSION

- **Lossless Audio Compression**
  - Removes ***redundant data***.
  - Resulting signal is ***same as original***-perfect reconstruction.
- **Lossy Audio Encoding**
  - Removes ***irrelevant data***.
  - Resulting signal ***is similar to original***.

AUDIO LEVEL COMPRESSION

- Audio level compression is also called as ***Dynamic Range compression.***
- Audio level compression in which the ***dynamic range (difference between loud and quiet)of an audio waveform is reduced.***
- This is used in **guitar effects racks, recording studios**, etc.
- **Types of Dynamic Range Compression:**

  - Downward compression.
  - Upward compression.

AUDIO LEVEL COMPRESSION

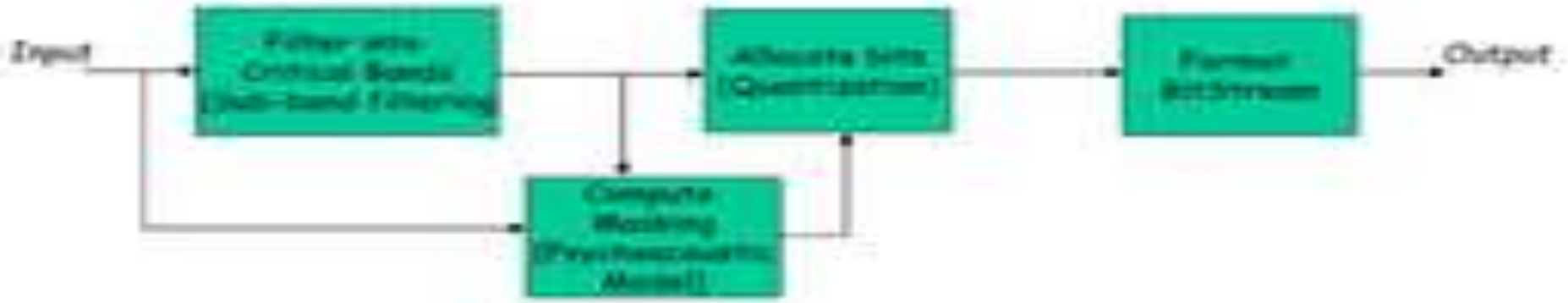
- **Downward compression**: It reduces loud sounds over a certain threshold while quiet sounds remain unaffected.
- **Upward compression**: It increases the loudness of sounds below a certain threshold while leaving louder sounds unaffected.

AUDIO COMPRESSION OPERATIONS

- Maintains the quality of sound the following operations is to be performed on audio signals.

- ❑ Quantization
- ❑ Splitting up the signal
- ❑ Psychoacoustic models/methods
- ❑ Frequency masking
- ❑ Temporal masking

MPEG Coding Algorithm



1. Use convolution filters to divide the audio signal (e.g., 48 kHz sound) into 32 frequency sub-bands. (*sub-band filtering*)
2. Determine amount of masking for each band caused by nearby band using the *psychoacoustic model*.
3. If the power in a band is below the masking threshold, don't encode it.
4. Otherwise, determine number of bits needed to represent the coefficient such that, the noise introduced by quantization is below the masking effect (Recall that one fewer bit of quantization introduces about 6 dB of noise).
5. Format bitstream

AUDIO COMPRESSION TECHNIQUES

- Audio compression can be ***used for speech or music***.
  - -For ***speech*** need to compress a ***64 kHz digital signal***,
  - -for ***music*** need to compress a ***1.411MHz signal***.
- Two categories of techniques are used for audio compression:

  - ❑Predictive encoding
  - ❑Perceptual encoding

PREDICTIVE ENCODING

- In predictive encoding, the ***differences between samples are encoded instead of encoding all the sampled values.***
- This compression is normally ***used for speech***.
- Several ***standards*** have been defined such as ***GSM*** (13kbps), ***G.729***(8 kbps), ***G.723.3***(6.4 or 5.3 kbps).

PERCEPTUAL ENCODING

- ***MP3***(M PEG audio layer 3).
- The most common compression technique used to create ***CD-quality audio*** is based on the perceptual encoding technique.
- This type of audio needs ***at least 1.411 Mbps***, which cannot be sent over the internet without compression.
- Algorithms used in this first transform the data from ***time domain to frequency domain,*** the operations are then performed to the data in the frequency domain. This technique is also called the ***frequency domain method***.
- It deals with ***two types of masking process*** that are ***frequency masking and temporal masking.***

PERCEPTUAL ENCODING

- In **frequency masking** , a loud sound in a frequency range can partially or totally mask a softer sound in another frequency range.
- In **temporal masking** , a loud sound can numb our ears for a short time even after the sound has stopped.
- MP3 uses these two phenomena to compress audio signals.
- The techniques ***analyses and divides the spectrum*** into several groups.

-There are certain frequency ranges that are ***totally masked*** to them ***ZERO bits are allocated***.

- -A ***fewer number of bits are allocated*** to those that are ***partially masked.***

- A ***larger number of bits are allocated*** to those that are ***not masked***

ADVANTAGES OF AUDIO COMPRESSION

- Faster transmission time.
- Reduced transmission costs.
- Smaller size.
- Reduced bandwidth due to smaller size.

DISADVANTAGES OF AUDIO COMPRESSION

- Compression can ***only be used if both the transmitting and receiving modems support the same compression procedure.***
- Needs ***processing both for encoding and decoding.***
- If ***a lossy*** compression method is used, the ***quality is reduced***.
- ***Lossless compression*** methods exists and they achieve about ***50% reduction in size***.

IMAGE COMPRESSION

The objective of image compression is **to reduce irrelevant and redundant image data** in order to be able to store or transmit data in an efficient form.

- Lossless image compression.
- Lossy image compression

TYPES

- **Lossless** image compression is a compression algorithm that ***allows the original image to be perfectly reconstructed from the original data***.
- **Lossy** image compression is a type of compression where a certain amount of information is discarded which means that some data are lost and hence the ***image cannot be decompressed with 100% originality***.

JPEG

- JPEG which stands for **Joint Photographic Experts Group** is a ***lossy compression algorithm for images***.
- A ***lossy compression*** scheme is a way to **inexactly represent** the data in the image, such that ***less memory is used*** yet the data appears to be very ***similar***.
- This is why JPEG look almost the same as the original images they were derived from most of the time, unless the ***quality is reduced significantly, in which case there will be visible differences***.

- The JPEG algorithm takes advantage of ***the fact that humans cannot see colours at high frequencies***.
- The high frequencies are the data points in the image that are ***eliminated during the compression***.
- JPEG compression also ***works best on images with smooth colour transitions.***

WHY JPEG COMPRESSION?

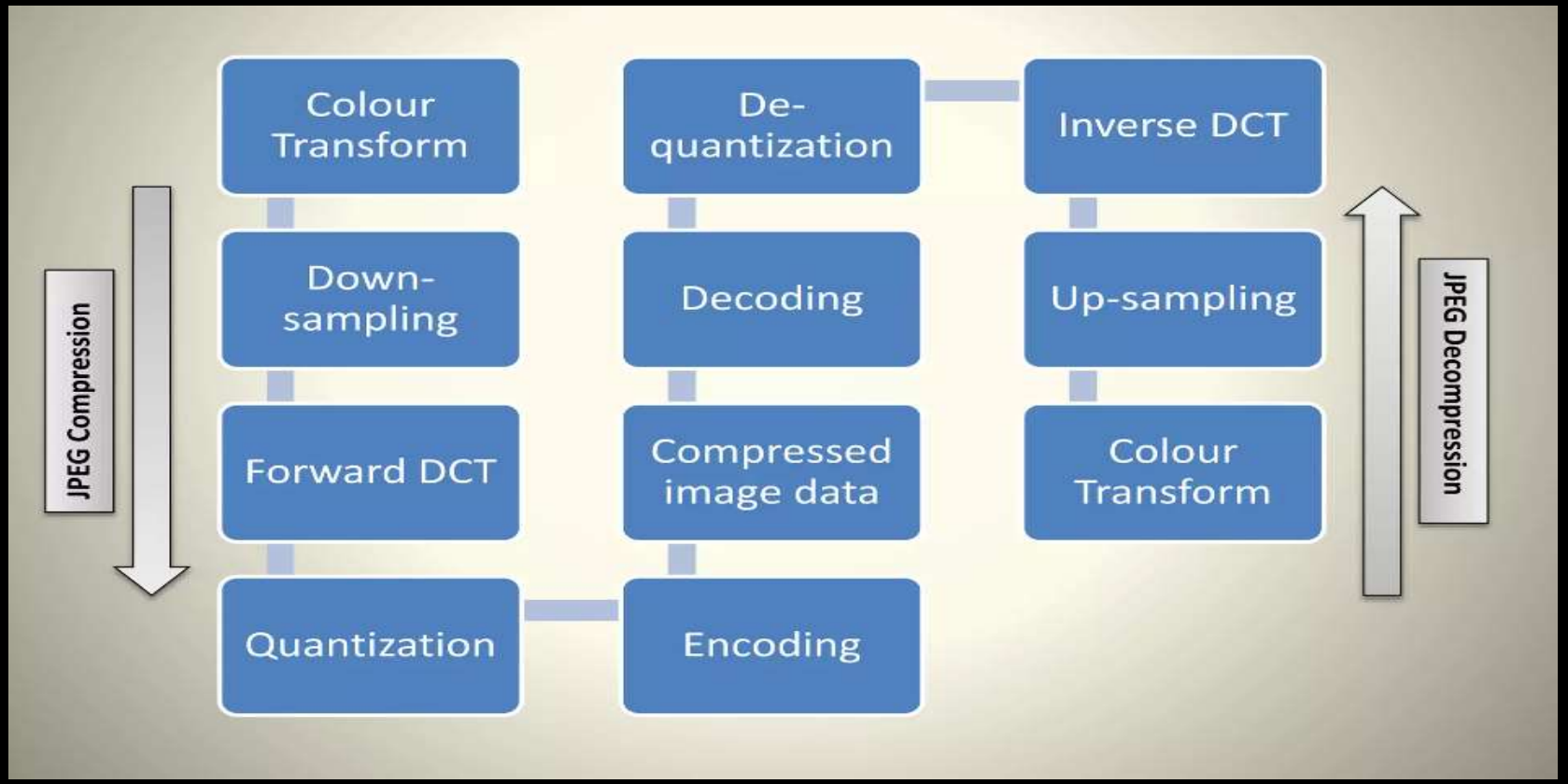
JPEG uses transform coding, it is largely based on the following observations:

- A large majority of useful image contents change relatively slowly across images, that is, it is unusual for intensity values to alter up and down several times in a small area.
- For example within an 8 x 8 image block. A translation of this fact into the spatial frequency domain, implies, generally, lower spatial frequency components contain more information than the high frequency components which often correspond to less useful details and noises.

JPEG COMPRESSION

- Experiments suggest that humans are more immune to loss of higher spatial frequency components than loss of lower frequency components. ***Human vision is insensitive to high frequency components.***

JPEG SCHEMATIC



STEPS

- **Splitting**: Split the image into 8×8 non-overlapping pixel blocks. If the image cannot be divided into 8-by -8 blocks, then can add in empty pixels around the edges, essentially zero -padding the image.
- **Colour Space Transform** from [R, G, B]to [Y, Cb, Cr] & Subsampling.
- **DCT:** take the Discrete Cosine Transform(DCT) of each 8-by -8 block.

STEPS

- **Quantization** : Quantize the DCT coefficient according to psycho visually tuned quantization tables.
- **Serialization** : by zigzag scaning pattern to exploit redundancy.
- **Vectoring** : Differential pulse code modulation(DPCM) on DC components.
- **Run length encoding**(RLE) on AC components
- **Entropy coding**:-run length coding-Huffman coding or arithmetic coding.

SPLITTING

The input image is divided into smaller blocks having 8 x 8 dimensions, summing up to 64 units in total. Each of these units is called a pixel, which is the smallest unit of any image.



Original image

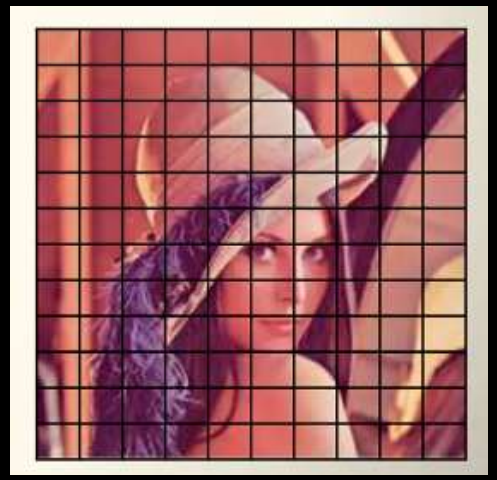
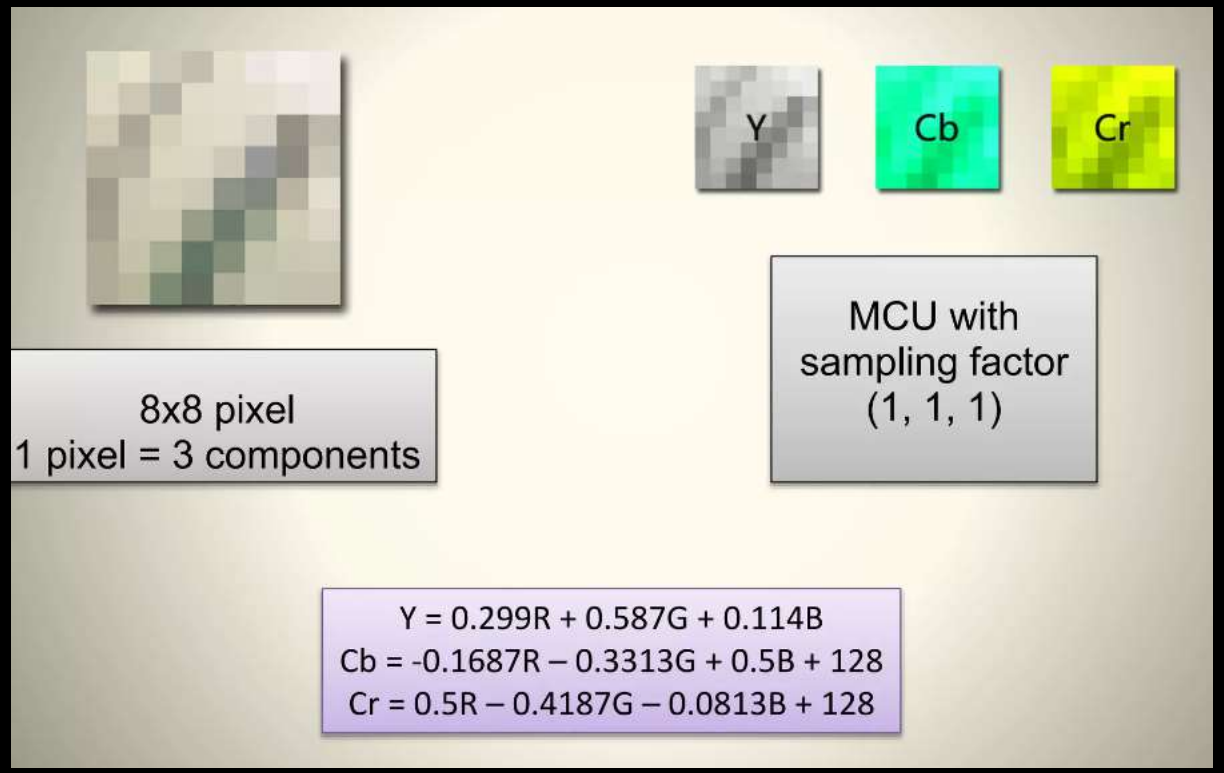


Image split in to multiple 8 x 8 pixel blocks

RGB TO YCBCR CONVERSION

- JPEG makes use of [Y, Cb, Cr] model instead of [R,G,B] model.
- The Y,Cb,Cr is used here because it seperates color into luminance and chrominance components which offers more efficient compression and less loss
  - JPEG uses [Y, Cb ,Cr] model
    Y=Brightness
    Cb= Colour blueness
    Cr=Colour redness

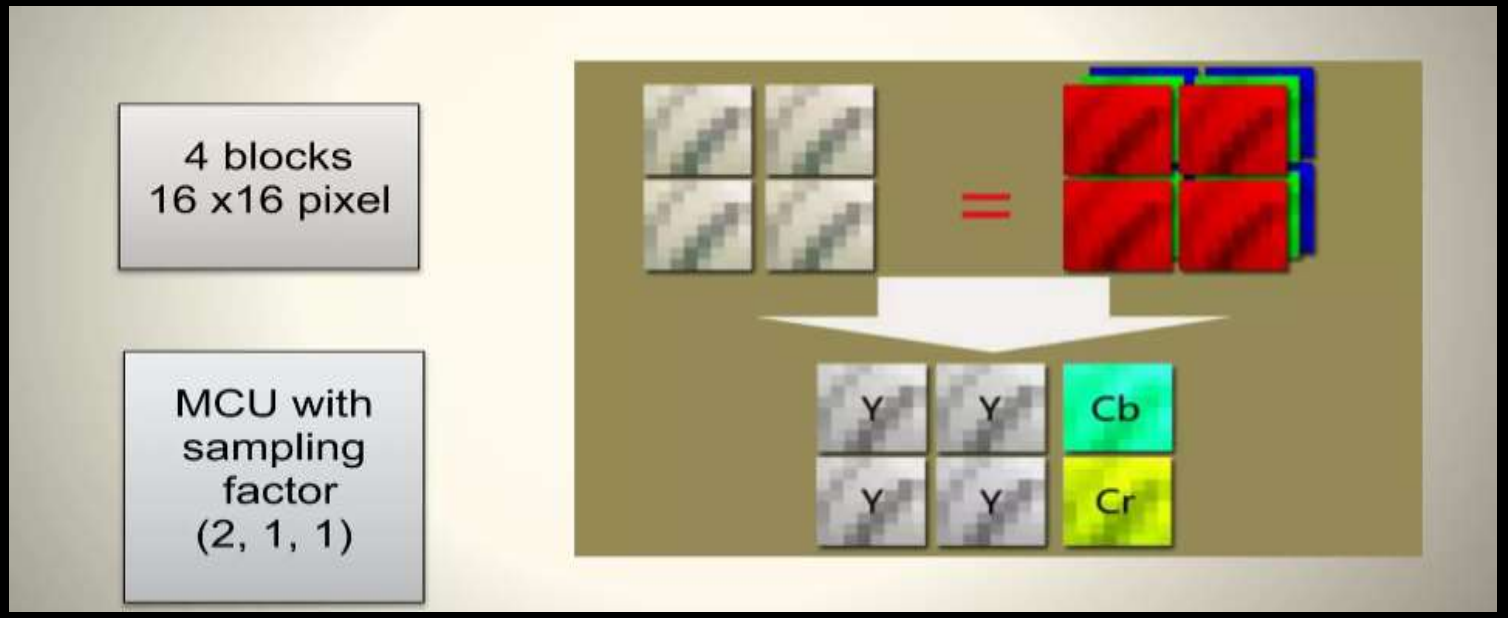
COLOUR CONVERSION



DOWN SAMPLING

➢ Y is taken for every pixel, and Cb , Cr are taken for a block of 2 x 2 pixels.

- This is done to redice spatial resolution for each Cb, Cr components



- MCU=Minimum Coded Unit(smallest unit that can be coded)
- Data size reduces to half.

FORWARD DCT

- DCT uses the cosine function, therefore not interacting with complex numbers at all.
- DCT converts the information contained in a block(8 x 8)of pixels from spatial domain to the frequency domain.
- If the input matrix is P(x,y) and the transformed matrix is F(u,v)or G(u,v) then the DCT for the 8 x 8 block is computed using the expression:
- F(0,0) is the DC components, rest are AC components

$$G_{u,v} = \frac{1}{4}\alpha(u)\alpha(v)\sum_{x=0}^{7}\sum_{y=0}^{7} g_{x,y}\cos\left[\frac{(2x+1)u\pi}{16}\right]\cos\left[\frac{(2y+1)v\pi}{16}\right]$$

QUANTIZATION

- Quantization reduce the number of bits per sample.

TYPES:

- **Uniform quantization :** q(u,v) is a constant.
- **Non-uniform quantization :** Custom quantization tables can be put in image/scan header and JPEG Standard defines two default quantization tables , one each for luminance and chrominance

QUANTIZATION FORMULA

Standard Formula: $\widehat{F}(u,v) = round\left(\frac{F(u,v)}{Q(u,v)}\right)$

- The quantization step is the major information losing step in JPEG compression.
- For non-uniform quantization , there are 2 psycho-visually tuned quantization tables each for luminance(Y) and chrominance(Cb ,Cr) components defined by JPEG standards.

QUANTIZATION TABLE

16	11	10	16	24	40	51	61
12	12	14	19	26	58	60	55
14	13	16	24	40	57	69	56
14	17	22	29	51	87	80	62
18	22	37	56	68	109	103	77
24	35	55	64	81	104	113	92
49	64	78	87	103	121	120	101
72	92	95	98	112	100	103	99

The Luminance Quantization table

17	18	24	47	99	99	99	99
18	21	26	66	99	99	99	99
24	26	56	99	99	99	99	99
47	66	99	99	99	99	99	99
99	99	99	99	99	99	99	99
99	99	99	99	99	99	99	99
99	99	99	99	99	99	99	99
99	99	99	99	99	99	99	99

The chrominance Quantization table

ZIG ZAG SCAN

- To group low frequency coefficients at the top of the vector and high frequency coefficients at the bottom.

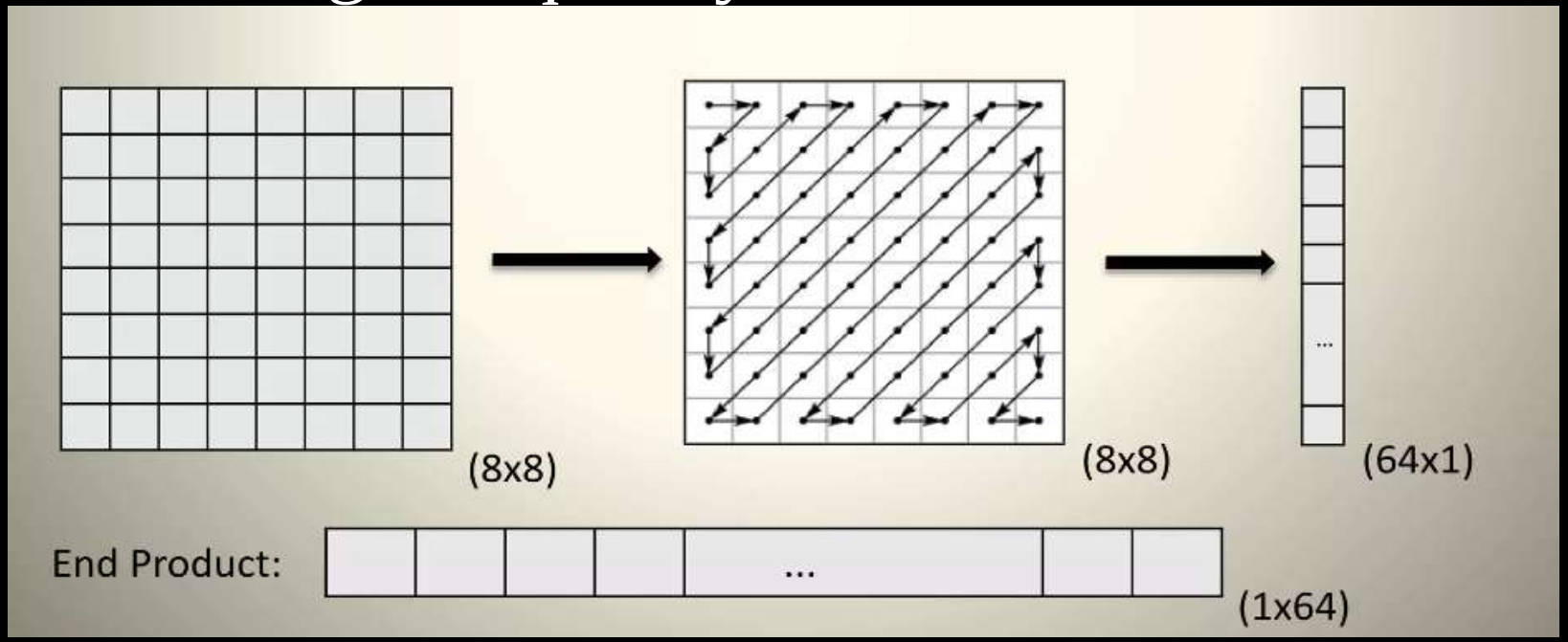


Figure: ZIG ZAG SCAN

VECTORING

- Differential pulse modulation (DPCM) is applied to the DC component.
- DC elements are used to define strength of colors
- The entropy coding operates on a 1-dimensional string of values(vector). However the output of the quantization matrix is a 2-D matrix and hence this has to be represented in 1-D form. This is known as vectoring.

ENCODING

- In the last stage, we apply to encode either run-length encoding or Huffman encoding. The main aim is to convert the image into text and by applying any encoding we convert it into binary form (0, 1) to compress the data.
- Run length encoding-Encoded as (skip , value) pairs, where skip is the number of zeros preceding a non-zero value in the quantized matrix, and value is the actual coded value of the non-zero component.

HUFFMAN CODING

- DC and AC components finally need to be represented by a smaller number of bits.
- Significant levels of compression can be obtained by replacing long strings of binary digits by a string of much shorter code words.

SIZE	Value	Code
0	0	---
1	-1,1	0,1
2	-3, -2, 2,3	00,01,10,11
3	-7,..., -4, 4,..., 7	000,..., 011, 100,...111
4	-15,..., -8, 8,..., 15	0000,..., 0111, 1000,..., 1111
.		.
.		.
11	-2047,..., -1024, 1024,... 2047	...

Table 1 : Huffman Table for DC components Value field

HUFFMAN CODING

SIZE	Code Length	Code
0	2	00
1	3	010
2	3	011
3	3	100
4	3	101
5	3	110
6	4	1110
7	5	11110
8	6	111110
9	7	1111110
10	8	11111110
11	9	111111110

Table 2: Huffman Table for DC component size field

MERITS OF IMAGE COMPRESSION

- Works with colour and grayscale images, but not with binary images.
- Up to 24 bits colour images
- Target photographic quality images(Unlike GIF).
- Suitable for many applications like satellite , medical , general photography etc.

DEMERITS

- Image ***may loss important details from the image*** because of lossy compression.
- JPEG is ***not efficient for image which contains texts*** , sharp lines or edges
- ***Quality of image is reduced(mainly*** for text based images).
- JPEG ***does not handle Black & White and motion pictures***.
- In JPEG ***layered image is not supported***.

.

THANK YOU

VIDEO COMPRESSION - MPEG

Video compression

- Aims in reducing the size of video

➢ There are two types of compression used in video encoding:

1. **Intra-frame compression** (also known as "intra-frame coding") and
2. **Inter-frame compression** (also known as "inter-frame coding")

Intra-frame compression

- It compresses each video frame independently, without taking into account the information from adjacent frames.
- This technique is also known as "frame-based compression" or "spatial compression".
- Each frame is divided into small blocks, and these blocks are encoded using a variety of algorithms such as Discrete Cosine Transform (DCT).
- It produces high-quality video with minimal artifacts.
- Best suited for video content with low motion or static images, such as photographs or text-based content.
- It requires more storage space compared to inter-frame compression because each frame is compressed independently.

Inter-frame compression

- It compresses video frames by taking into account the information from adjacent frames.
- This technique is also known as "motion-based compression" or "temporal compression".
- In inter-frame compression, the video frames are divided into groups of frames called "GOP" (Group of Pictures), and each GOP contains one or more I-frames, followed by one or more P-frames and B-frames.

Lossy and Lossless compression

Lossy compression :

- **Achieve higher compression ratios by selectively discarding some visual data.**
- **This results in smaller file sizes but comes with a trade-off in a slight reduction in quality.**
- **Commonly used for streaming and distribution as they effectively balance file size and visual quality**

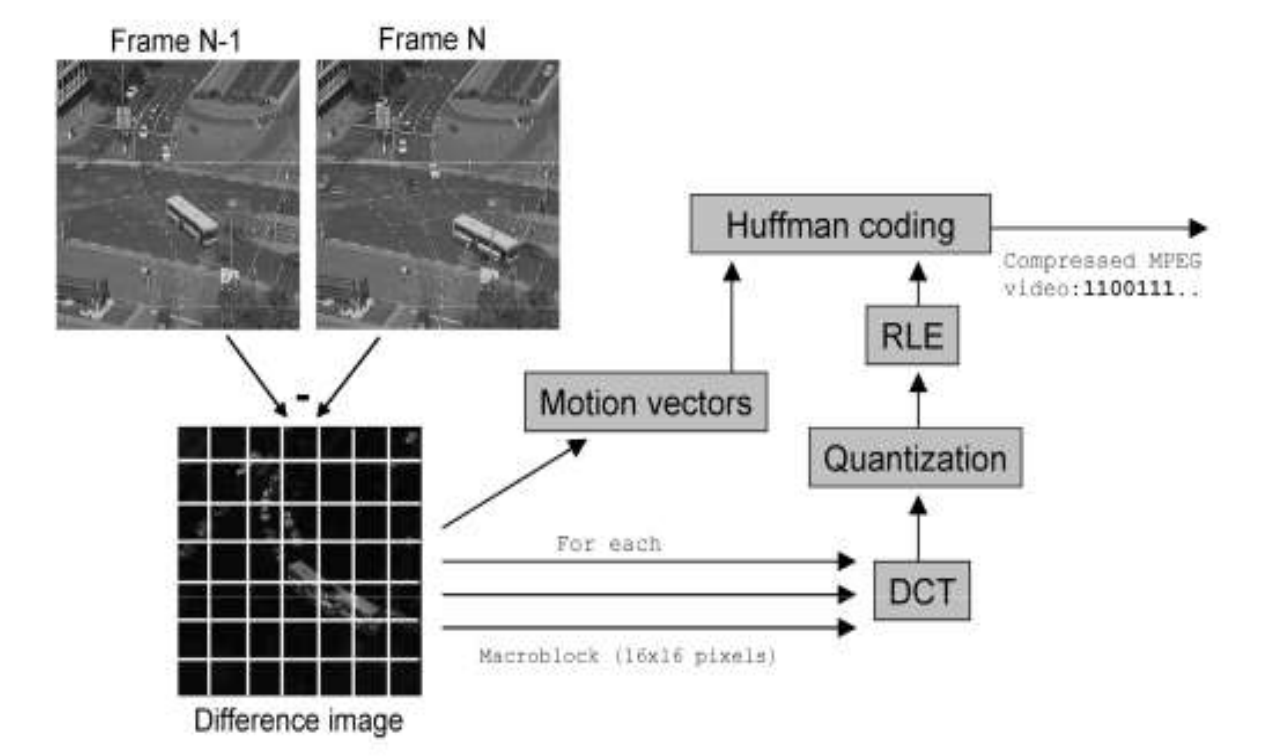
Lossless compression:

- **Preserve the entire video quality without losing information.**
- **Suitable for situations where maintaining fidelity to the source material is essential, such as in video editing or archival purposes.**
- **Result in larger file sizes compared to lossy codecs.**

MPEG

- MPEG (Moving Picture Experts Group) is a method for video compression
- It is a lossy video compression technique.
- MPEG file will have a .mpeg file extension.
- Some of the released version of MPEG are: MPEG-1, MPEG-2, MPEG-3, MPEG-4, MPEG-7, MPEG-21 and more
- Apply similar technique as JPEG to individual image.

MPEG compression steps :



MPEG compression steps: (Intra-frame mode)

1. Preliminary scaling and color conversion
2. Color subsampling
3. Discrete Cosine Transform (DCT)
4. Quantization
5. Run Length Encoding
6. Entropy Coding – Huffman Coding

1. PRELIMINARY SCALING AND COLOR CONVERSION:

- Each frame is converted into a standardized representation known as Source Input Format (SIF)
- Color information is translated into a scheme known as YUV.

2. Color subsampling:

- Brightness is the dominant component of color seen by the human eye, while hue is less important.
- The algorithm is able to reduce the hue information by about 75% with little effect on subjective fidelity.

3. Discrete cosine transformation (DCT):

- This process maps each 8 * 8 block of points (pixels) into a set of numbers similar to a Fourier transform of the block.
- The DCT provides a frequency domain representation of the image.

4. Quantization:

- The DCT values are quantized into a finite number of possible values .
- The more quantization levels that are used, the greater the picture fidelity, but the less the amount of compression.

5. Run-length encoding:

The quantized DCT values are represented using a run-length encoding technique.

6. Huffman coding:

- A lossless compression technique that assigns the most common bit sequences from the preceding step to symbols that are as short as possible.

- **For inter-frame mode, similar blocks of pixels common to two or more successive frames are replaced by a pointer that references one of the blocks.**

Motion Compensation

- **The foundation of MPEG interframe compression is motion compensation.**

- **The idea behind motion compensation is that a portion of an image in one frame will be the same as or very similar to an equal-sized portion in a nearby frame.**

- **The MPEG scheme makes use of two forms of motion compensation:**
  Prediction and Interpolation.

Prediction

- MPEG makes use of blocks of 16 * 16 pixels, called macroblocks (in contrast to the smaller 8 * 8 blocks used in the DCT coding), for purposes of motion compensation.
- A frame processed in prediction mode is divided into its macroblocks, each of which is encoded separately.
- The encoding is done with reference to an anchor frame that precedes the current frame.
- Each macroblock in the current frame is to be represented by a pointer, called the motion vector, to that macroblock in the anchor frame that most closely matches this macroblock.
- The motion vector gives the displacement of the macroblock in the current frame with respect to its match in the anchor frame

- **The motion vector for a macroblock is obtained by minimizing a cost function that measures the difference between a macroblock and each predictor candidate.**

The calculation can be expressed as

$$\underset{m \in M}{\text{MIN}} \left[\sum_{(x,y) \in B_i} C[I_c(x, y) - I_r((x, y) + m)] \right]$$

where

B_i = a macroblock in the current frame I_c

m = the displacement vector with respect to the reference frame I_r

M = the search range in the reference frame

C = cost function

Interpolation

- MPEG allows some video frames to be encoded using two reference frames, one in the past and one in the future. This approach, called bidirectional interpolation.
- Each macroblock can be encoded using a block from the preceding frame (forward prediction), the following frame (backward prediction), or one block from each reference frame (averaging), whichever gives the minimum error matrix.

Mode	Predictor
Forward predicted	$\hat{I}_1(z) = \hat{I}_0(z + M_{01})$
Backward predicted	$\hat{I}_1(z) = \hat{I}_2(z + M_{21})$
Average	$\hat{I}_1(z) = \frac{\hat{I}_0(z + M_{01}) + \hat{I}_2(z + M_{21})}{2}$

Note: z = the vector (x, y).

Frame Ordering

Three types of frames are defined in MPEG:

- **Intraframe (I)**: Encoded in JPEG style as an independent still image
- **Predicted (P)**: Encoded with reference to the preceding anchor frame.
- **Bidirectional interpolated (B)**: Encoded with reference to the preceding and the following anchor frames

The rules for encoding are as follows:

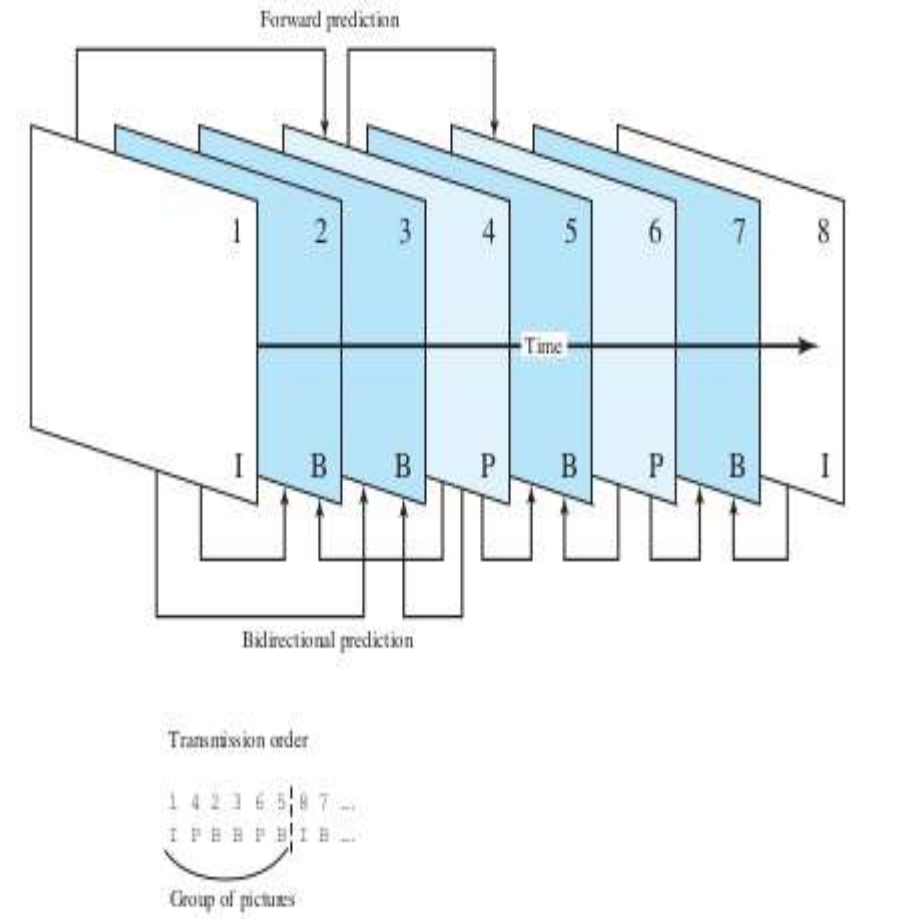
- **Each I frame is encoded using intraframe coding only.**
- **Each P frame is encoded based on the most recent preceding I or P frame, whichever is closest.**
- **Each B frame is encoded with the closest preceding and following I or P frames.**

➢ **Picture frames in MPEG are organized into groups.**

➢ **Each group consists of a single I frame followed by a number of P frames and B frames.**

In **the figure ,**

- **The first six frames form a group.**
- **Frame 4 is stored after frame 1, because it is used as the forward prediction frame for B frames 2 and 3.**
- **Frames 5 and 6 are interchanged for the same reason.**
- **B frame 7 is recorded as part of the next group because it is encoded after I frame 8.**



Advantages of MPEG:

- **High compression efficiency** − Highly efficient compression standard and can significantly reduce the file size of audio and video files while maintaining good quality.
- **Widely supported** − MPEG is a widely used and well-established audio and video format, and it is supported by a wide range of media players, video editors, and other software.
- **Good quality** − While MPEG uses lossy compression, it can still produce good quality audio and video at moderate to high compression levels.
- **Flexible** − The degree of compression used in an MPEG file can be adjusted, allowing you to choose the balance between file size and quality.

- **Versatile** − MPEG can be used with a wide range of audio and video types, including music, movies, television shows, and other types of multimedia content.
- **Streamable** − MPEG files can be streamed over the internet, making it easy to deliver audio and video content to a wide audience.
- **Scalable** − MPEG supports scalable coding

Disadvantages of MPEG:

- **Lossy compression** − Because MPEG uses lossy compression, some data is lost when the audio or video is compressed. This can result in some loss of quality, particularly at higher levels of compression.
- **Limited color depth** − Some versions of MPEG have a limited color depth and can only support 8 bits per channel. This can result in visible banding or other artifacts in videos with high color gradations or smooth color transitions.

- **Non-ideal for text and graphics** − MPEG is not well suited for video with sharp transitions, high-contrast text, or graphics with hard edges. These types of video can appear pixelated or jagged when saved as MPEG.
- **Complexity** − The MPEG standards are complex and require specialized software and hardware to encode and decode audio and video.
- **Patent fees** − Some MPEG standards are covered by patents, which may require the payment of licensing fees to use the technology.
- **Compatibility issues** − Some older devices and software may not support newer versions of the MPEG standard.